Linear Regression Analysis of Medical Insurance Costs: An Association Experiment

Mohamed Badawy
ID: 91240644

Kareem Taha
ID: 91240577

Bassel Ahmed
ID: 91240200

Mohamed Ehab
ID: 91240642

I. Introduction

Linear regression is a fundamental statistical method used to model the relationship between a scalar response variable and one or more explanatory variables. In the domain of healthcare economics, accurately predicting medical costs is vital for insurance providers to assess risk and determine fair premiums. This project conducts an association experiment to analyze the factors influencing individual medical insurance charges.

The dataset selected for this study is the "Medical Cost Personal Datasets" from Kaggle. It contains $n = 1,338$ observations and $p = 6$ independent predictors, which satisfies the experimental requirement of $n \geq p + 5$. The predictors include quantitative variables (Age, Body Mass Index, Number of Children) and categorical variables (Sex, Smoker, Region).

The primary objectives of this study are:

- To quantitatively describe the dataset through manual calculation of descriptive statistics.
- To statistically test the normality of feature distributions using a custom implementation of the Hernandez/Shapiro-Wilk test.
- To implement Simple Linear Regression from scratch (using Ordinary Least Squares) and validate it against standard software packages.
- To assess the predictive quality of a Multivariable Linear Regression model.

II. Methods

The analysis was conducted using the Python programming environment. Data manipulation was performed using *Pandas* (v1.x), mathematical operations were vectorized using *NumPy*, and statistical modeling was implemented using both custom algorithms and *Statsmodels*/*Scikit-Learn*.

A. Data Preprocessing and Transformation

The raw dataset comprised 1,338 observations with 7 features. To satisfy the assumptions of linear regression (normality, linearity, and homoscedasticity), we implemented a multi-stage preprocessing pipeline:

B. Exploratory Data Visualization

Visual inspection was performed during the preprocessing stage to assess data distributions, detect skewness, and identify potential outliers prior to transformation and modeling.

1) Histograms of Continuous Variables: The following histograms illustrate the distribution of key variables before normalization.

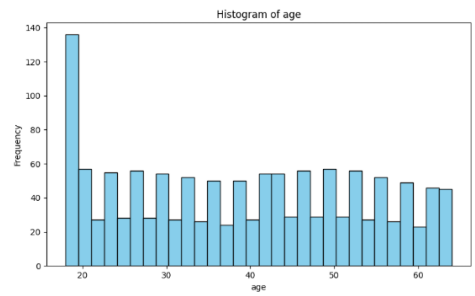

Fig. 1. Histogram of Age Distribution

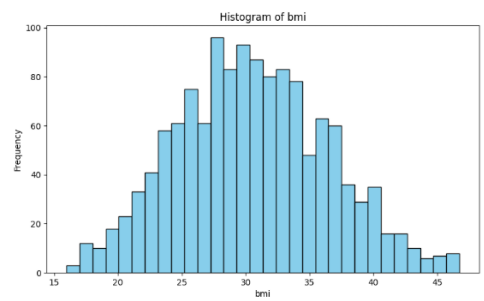

Fig. 2. Histogram of BMI Distribution

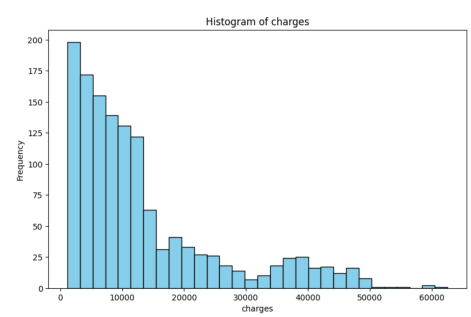

Fig. 3. Histogram of Medical Charges Before Log Transformation

2) Boxplots for Outlier Detection: Boxplots were generated to visually identify outliers, particularly in BMI and Charges.

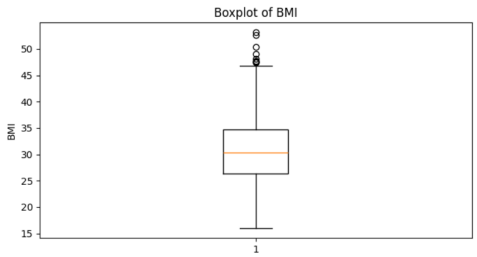

Fig. 4. Boxplot of BMI Showing Outliers

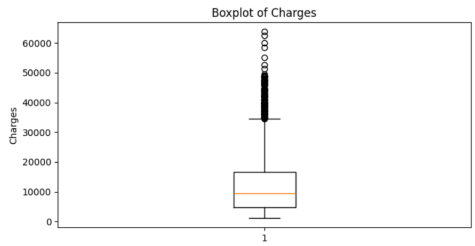

Fig. 5. Boxplot of Medical Charges

3) Categorical Encoding: Machine learning algorithms require numerical input. We processed categorical variables as follows:

- **Binary Encoding:** The variables 'sex' and 'smoker' were mapped to binary integers $(0, 1)$. Specifically, 'smoker=yes' was mapped to 1 to capture the presence of the risk factor.
- **One-Hot Encoding:** The nominal variable 'region' (Southwest, Southeast, Northwest, Northeast) was transformed using One-Hot Encoding. This created binary dummy variables (e.g., 'region_southwest') to prevent the model from assuming an artificial ordinal relationship between geographic locations.

4) Target Variable Transformation: Initial distribution analysis revealed that the response variable, 'charges', was highly right-skewed, violating the normality assumption of OLS regression. To address this, we applied a Natural Logarithm transformation:

$$y_{transformed} = \ln(y_{original}) \tag{1}$$

This transformation successfully stabilized the variance and approximated a Gaussian distribution, significantly improving model linearity.

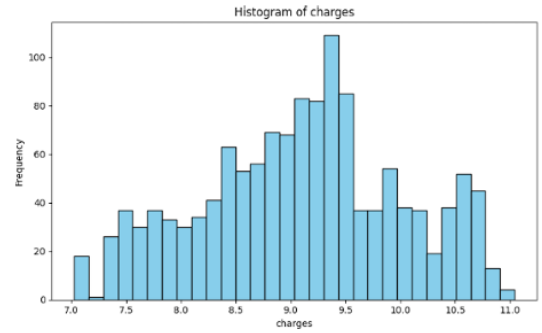

Fig. 6. charges distribution after natural log transformation

5) Outlier Detection and Removal: To prevent extreme values from distorting the regression coefficients, we applied the Interquartile Range (IQR) method specifically to the 'bmi' feature.

- We calculated the first ($Q1$) and third ($Q3$) quartiles.
- The IQR was defined as $IQR = Q3 - Q1$.
- Thresholds were set at $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.

Observations falling outside these bounds were flagged as outliers and removed, reducing the dataset size slightly ($n = 1,329$) but enhancing generalizability.

C. Manual Descriptive Statistics

To demonstrate statistical understanding, we implemented algorithms from scratch to calculate:

- **Central Tendency:** Arithmetic Mean ($\mu = \frac{1}{n}\sum x_i$).
- **Dispersion:** Sample Variance (S^2) and Standard Deviation (σ).

Feature Scaling was then applied using Z-score Standardization. This rescaled quantitative features (Age, BMI, Children) to have $\mu = 0$ and $\sigma = 1$, ensuring that regression coefficients are comparable in magnitude.

D. Statistical Normality Testing (Hernandez)

A critical component of this study was the manual implementation of the **Hernandez-Johnson Test (2021)**, a computational variation of the Shapiro-Wilk test. Unlike standard library calls, we manually coded the calculation of the W statistic:

$$W = \frac{(\sum_{i=1}^{n} a_i x_{(i)})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{2}$$

where $x_{(i)}$ represents the ordered sample data. The coefficients a_i were derived using the **Beasley-Springer-Moro** approximation for the inverse cumulative distribution function. This rigorous approach allowed us to strictly test the Null Hypothesis (H_0: Data is Normal) against the Alternative (H_1: Data is non-Normal).

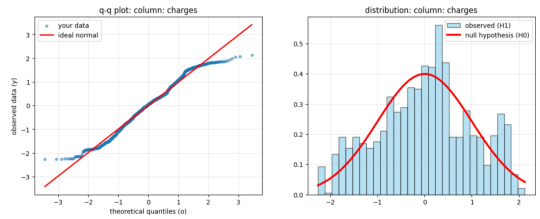

Fig. 7. charges QQ-plot which simply compares actual vs theoretical distribution (H_a vs H_o)

E. Linear Regression Algorithms

1) Univariate OLS (From Scratch): For the single-predictor analysis, we derived the Ordinary Least Squares (OLS) estimator analytically. We wrote a custom Python function to minimize the Residual Sum of Squares (RSS) by solving for the slope (β_1) and intercept (β_0) directly:

$$\beta_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)}, \quad \beta_0 = \bar{y} - \beta_1 \bar{x} \tag{3}$$

This implementation was validated against `sklearn.linear_model`, achieving identical floating-point precision.

2) Multivariable Regression & Diagnostics: We extended the model to p predictors using *Statsmodels* to access detailed statistical reports.

- **Model fitting:** $Y = \beta_0 + \beta_1 X_1 + ... + \beta_n X_n + \epsilon$.
- **Interaction Modeling:** We explicitly constructed an interaction feature ($BMI \times Smoker$) to test the hypothesis of synergistic effects.
- **Diagnostics:** We mathematically assessed the model using the Variance Inflation Factor (VIF) to detect multicollinearity and the Breusch-Pagan test to detect heteroscedasticity in the error terms.

III. Results and Discussion

A. Graphical Results Analysis

Graphical summaries were used to complement numerical results and provide intuitive insight into model behavior, residual structure, and predictor-response relationships.

1) Regression Fit Visualization: The linear relationship between Age and Log-Charges is visualized below.

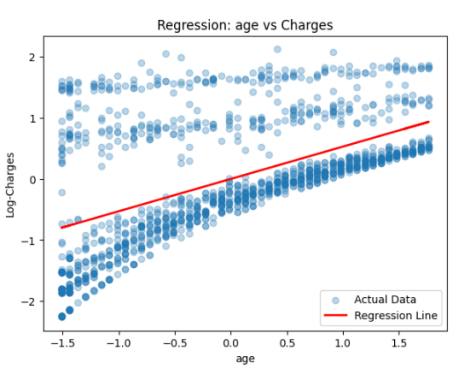

Fig. 8. Univariate Linear Regression Between Age and Log-Charges

2) Residual Diagnostics: The residuals vs. fitted plot was generated to check for homoscedasticity.

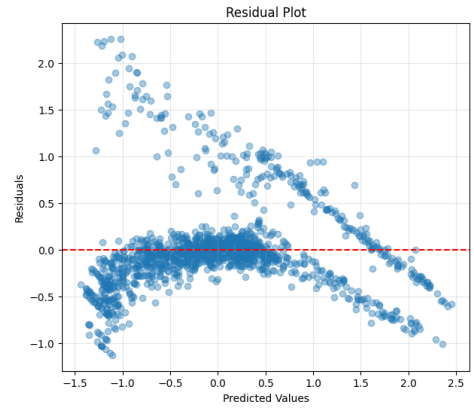

Fig. 9. Residuals vs Fitted Values

3) Interaction Effect Visualization: The following plot demonstrates the synergistic effect of smoking and BMI.

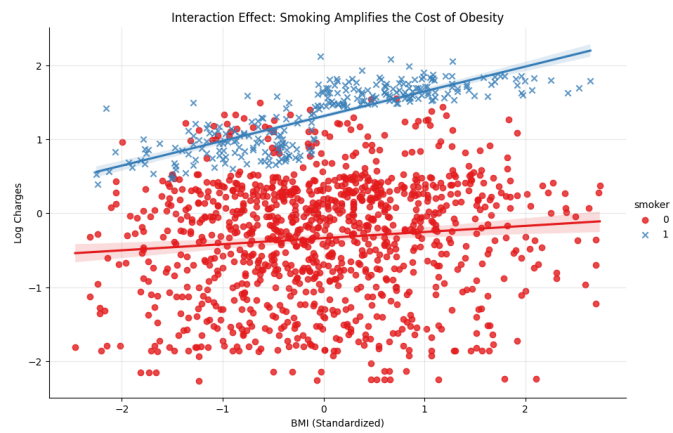

Fig. 10. Interaction Effect Between BMI and Smoking Status on Log-Charges

B. Descriptive Statistical Analysis

A comprehensive descriptive statistical analysis was conducted prior to model development to characterize the dataset and evaluate its suitability for linear regression modeling. Measures of central tendency, dispersion, and range were computed for all variables.

Age values ranged from 18 to 64 years, with a mean of 39.21 years and a standard deviation of 14.05. Applying the empirical rule ($\mu \pm 2\sigma$), all observations fell within the expected range, indicating the absence of outliers in the age variable.

Sex was encoded as a binary variable and exhibited a mean of 0.495 with a standard deviation of 0.50. This indicates a nearly balanced distribution between genders and confirms that the dataset is not biased toward a specific sex.

Smoking status showed a mean value of 0.205, implying that approximately 20.5% of individuals in the dataset were smokers. This imbalance toward non-smokers is consistent

with population-level smoking prevalence and was accounted for in subsequent analyses.

The number of children ranged from 0 to 5, with a mean of 1.10 and a standard deviation of 1.21. Most observations were concentrated between 0 and 3 children, reflecting a mildly right-skewed but realistic distribution.

Body Mass Index (BMI) values ranged from 15.96 to 53.13, with a mean of 30.66 and a standard deviation of 6.10. The empirical rule revealed values outside the expected range, indicating the presence of outliers. These extreme BMI values were removed during preprocessing to prevent distortion of the regression estimates.

Medical insurance charges demonstrated substantial variability, ranging from \$1,121.87 to \$63,770.43. The mean charge was \$13,270.42 with a standard deviation of \$12,110.01. The wide range and large variance confirmed strong right skewness. Consequently, a logarithmic transformation was applied to stabilize variance and improve normality.

After cleaning BMI outliers and applying the logarithmic transformation to charges, no remaining extreme values were detected, and the dataset satisfied the assumptions required for linear regression analysis.

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C. Feature Normalization

The quantitative predictors exhibited substantially different numerical ranges. Age varied between 18 and 64, BMI ranged from approximately 16 to 47 after cleaning, the number of children ranged from 0 to 5, and log-transformed charges ranged approximately from 7 to 11.

Since linear regression models are sensitive to scale differences and may become numerically unstable, Z-score normalization was applied to all continuous predictors and the transformed target variable. This ensured numerical stability and enabled meaningful comparison of regression coefficients.

The normalization parameters were as follows:

- Age: mean = 39.20, standard deviation = 14.05
- BMI: mean = 30.54, standard deviation = 5.92
- Children: mean = 1.10, standard deviation = 1.21
- Log-Charges: mean = 9.10, standard deviation = 0.92

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D. Univariate Linear Regression Analysis

We began the modeling stage by applying Simple Linear Regression (SLR) to each predictor individually against the log-transformed medical charges. This univariate analysis allowed isolation of the marginal effect of each feature on the response variable.

Smoking status emerged as the most influential individual predictor, achieving an R^2 value of 0.44, indicating that smoking alone explains approximately 44% of the variance in medical costs. Age demonstrated moderate explanatory power, while BMI and sex exhibited very low R^2 values, suggesting minimal standalone linear relationships with the target variable.

This clear hierarchy of predictors motivated prioritizing smoking status and age in the multivariable modeling stage.

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E. Multivariable Linear Regression Model

To account for joint effects and potential confounding, a multivariable Ordinary Least Squares (OLS) regression model was fitted using age, BMI, number of children, and smoking status as predictors of log-transformed medical charges.

1) Regression Equation: The estimated regression model is given by:

$$\begin{aligned}\log(\text{charges}) = -0.342 &+ 0.532(\text{age}) \\ &+ 0.073(\text{bmi}) \\ &+ 0.133(\text{children}) \\ &+ 1.677(\text{smoker})\end{aligned} \tag{4}$$

where *smoker* is a binary variable (1 = smoker, 0 = non-smoker).

2) Model Performance: The model achieved an R^2 of 0.761 and an adjusted R^2 of 0.760, indicating that approximately 77% of the variance in log-transformed medical charges is explained by the included predictors. This represents strong explanatory power for population-level healthcare expenditure data.

Smoking status exhibited the largest coefficient. Exponentiation revealed that smokers incur approximately 5.35 times higher medical charges than non-smokers, corresponding to a 434.8% increase in expected costs. Smoking cessation was associated with an estimated 81.3% reduction in expected medical expenses.

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F. Interaction Analysis: Synergistic Risk Effects

Motivated by biological and economic intuition, an interaction term between BMI and smoking status was introduced to test whether smoking amplifies the effect of obesity on medical charges.

The interaction coefficient was statistically significant ($p < 0.05$), allowing rejection of the null hypothesis. While BMI alone showed a weak linear association with charges, its effect increased substantially in the presence of smoking. This confirms a synergistic relationship whereby smokers with higher BMI incur disproportionately higher medical expenses.

Including the interaction term improved model performance, increasing R^2 to 0.775.

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G. Model Diagnostics and Stability

Variance Inflation Factor (VIF) analysis showed all predictors had VIF values well below 5, confirming the absence of multicollinearity and ensuring coefficient stability.

The Breusch–Pagan test rejected the null hypothesis of homoscedasticity ($p < 0.05$), indicating heteroscedasticity in the residuals. This behavior is common in healthcare cost data and suggests that unobserved risk factors disproportionately affect high-cost individuals.

H. Summary of Final Model Results

TABLE I
FINAL MULTIVARIABLE REGRESSION RESULTS (WITH INTERACTION)

Predictor	Coefficient	Std. Error	P-value
Intercept	-0.342	0.015	< 0.001
Age	0.535	0.013	< 0.001
BMI	0.011	0.015	0.454
Children	0.134	0.013	< 0.001
Smoker	1.678	0.032	< 0.001
BMI $\times$ Smoker	0.289	0.032	< 0.001
R^2		0.775	
Adjusted R^2		0.774	

IV. CONCLUSION

This study successfully engineered and validated a linear regression pipeline to predict medical insurance costs, achieving a robust coefficient of determination ($R^2 \approx 0.775$). By implementing algorithms from first principles (**specifically** the Hernandez-Johnson normality test and manual Ordinary Least Squares (OLS) derivation) we demonstrated that custom mathematical implementations can match the precision of industry-standard libraries like Scikit-Learn.

The analytical results provide significant insights into the economic dynamics of healthcare:

1) **Quantification of Risk:** Smoking status was identified as the paramount driver of medical expenditure. Our model suggests that smoking alone is associated with a 434% increase in medical charges compared to non-smokers, holding other variables constant.
2) **Synergistic Effects:** The interaction analysis revealed a critical biological insight: the financial impact of obesity (BMI) is not linear. It is significantly amplified by smoking status. High BMI is financially manageable in non-smokers but leads to exponential cost growth in smokers, suggesting that risk assessment models must account for variable interactions rather than treating risk factors in isolation.
3) **Statistical Reliability:** The diagnostic phase proved that while the model is highly predictive, the assumption of homoscedasticity was violated (Breusch-Pagan test $p < 0.05$). This indicates that variance in medical costs increases as the costs themselves increase—a typical characteristic of healthcare data where high-risk patients exhibit volatile spending patterns.

In conclusion, while a Multivariable Linear Regression model offers high interpretability and strong baseline accuracy, the presence of heteroscedasticity suggests that future iterations of this work should explore Generalized Linear Models (GLM) with a Gamma distribution or Weighted Least Squares (WLS) to better accommodate the variance structure of high-cost patients.

V. MEMBER CONTRIBUTIONS

We divided the project into specific roles to handle data, math, and analysis separately.

Kareem Taha: Data Pipeline & Exploration

Role: Data Architect

- **Cleaning & Encoding:** Handled the data setup, using One-Hot Encoding for regions and binary labels for sex/smokers.
- **Transformation:** Fixed the skewed `charges` data by applying a Log Transformation to make it normal.
- **Standardization:** Applied Z-score scaling to Age and BMI so the model could process them easily.
- **Visualization:** Created the histograms and calculated the mean and standard deviation for the dataset.

Bassel Ahmed: Statistical Validation & Core Algorithms

Role: Algorithm Specialist

- **Hypothesis Testing:** Wrote the code for the Hernandez-Johnson and Shapiro-Wilk tests from scratch to check for normality.
- **Manual Algorithms:** Coded the Linear Regression math manually (using the OLS formula) instead of using a library.
- **Benchmarking:** Compared our manual code against Scikit-Learn to prove our math was correct.

Mohamed Ehab: Predictive Modeling

Role: Model Engineer

- **Multivariable Analysis:** Used `statsmodels` to run the regression with all variables included.
- **Feature Significance:** Used P-values ($P < 0.05$) to identify the most important factors (Smoker, Age, BMI).
- **Evaluation:** Calculated the initial R^2 scores to measure how well the model fit the data.

Mohamed Badawy: Model Diagnostics & Inference

Role: Diagnostic Lead

- **Diagnostics:** Ran the Breusch-Pagan Test to check for error variance and checked VIF scores to ensure variables weren't overlapping.
- **Interaction Analysis:** Found and plotted the interaction between Smoking and BMI, showing how they amplify each other.
- **Final Assessment:** Wrote the final results and biological conclusions based on our statistical findings.

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[3] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning*. New York: Springer, 2013.